

assignment_03_key

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Introduction

This analysis examines whether crayfish chelae (claw) length differs between upstream and downstream locations relative to waterfalls across six different rivers. We hypothesize that downstream crayfish will have larger chelae due to increased food availability and different selective pressures.

Study Design: - Fixed effect: Position (Upstream vs Downstream)

- Random effect: River (6 rivers with extreme variation) - Response variable: Chelae length (mm) - Multiple individuals sampled at each position within each river

This analysis demonstrates why mixed-effects models are superior to averaging data and using simple one-way ANOVA.

Load Required Packages

```
library(tidyverse)    # Data manipulation and visualization
```

```
— Attaching core tidyverse packages ————— tidyverse 2.0.0 —
✓ dplyr      1.1.4      ✓ readr      2.1.5
✓ forcats    1.0.0      ✓ stringr    1.5.1
✓ ggplot2    3.5.2      ✓ tibble     3.3.0
✓ lubridate  1.9.4      ✓ tidyr      1.3.1
✓ purrr      1.1.0
— Conflicts ————— tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lme4)          # Mixed-effects models
```

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

```
library(car)           # ANOVA and assumption testing
```

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

```
recode
```

The following object is masked from 'package:purrr':

```
some
```

```
library(DHARMA)      # Residual diagnostics for GLMMs
```

This is DHARMA 0.4.7. For overview type '?DHARMA'. For recent changes, type news(package = 'DHARMA')

```
library(emmeans)     # Post-hoc comparisons and contrasts
```

Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'

```
library(broom.mixed) # Tidy model outputs  
library(ggthemes)    # Additional ggplot themes  
library(patchwork)   # Combine plots  
library(lmerTest)    # F-tests for mixed models
```

Attaching package: 'lmerTest'

The following object is masked from 'package:lme4':

```
lmer
```

The following object is masked from 'package:stats':

```
step
```

```
# Set theme for plots  
theme_set(theme_clean() +  
  theme(strip.background = element_rect(fill = "white", color = "black"),  
    panel.border = element_rect(color = "black", fill = NA)))
```

Generate Example Data with Extreme River Variation

```
set.seed(42) # For reproducible results  
  
# Define study parameters  
rivers <- paste0("River_", LETTERS[1:6])  
positions <- c("Upstream", "Downstream")  
n_per_group <- 15 # individuals per position per river  
  
# Generate realistic data with EXTREME river variation and downstream effect  
generate_crayfish_data <- function() {  
  
  # Create base data structure
```

```

data <- expand_grid(
  river = rivers,
  position = positions,
  individual = 1:n_per_group
) %>%
mutate(
  # Create unique individual IDs
  crayfish_id = paste(river, position, individual, sep = "_"),

  # EXTREME river-specific random effects (major differences between rivers)
  river_effect = case_when(
    river == "River_A" ~ rnorm(n(), 3.5, 0.8), # Very large crayfish
    river == "River_B" ~ rnorm(n(), -4.2, 0.8), # Very small crayfish
    river == "River_C" ~ rnorm(n(), 5.8, 0.8), # Extremely large crayfish
    river == "River_D" ~ rnorm(n(), -3.8, 0.8), # Very small crayfish
    river == "River_E" ~ rnorm(n(), 2.1, 0.8), # Large crayfish
    river == "River_F" ~ rnorm(n(), -1.5, 0.8) # Small crayfish
  ),

  # Position effect (downstream larger) - consistent across rivers
  position_effect = ifelse(position == "Downstream", 2.8, 0),

  # Individual variation
  individual_variation = rnorm(n(), 0, 1.0),

  # Calculate chelae length
  chelae_length = 12 + position_effect + river_effect + individual_variation,

  # Ensure no negative values
  chelae_length = pmax(chelae_length, 3)
) %>%
select(crayfish_id, river, position, chelae_length)

return(data)
}

# Generate the dataset
crayfish_data <- generate_crayfish_data()

# Display structure and summary
str(crayfish_data)

```

```

tibble [180 × 4] (S3: tbl_df/tbl/data.frame)
 $ crayfish_id   : chr [1:180] "River_A_Upstream_1" "River_A_Upstream_2" "River_A_Upstream_3"
"River_A_Upstream_4" ...
 $ river         : chr [1:180] "River_A" "River_A" "River_A" "River_A" ...
 $ position      : chr [1:180] "Upstream" "Upstream" "Upstream" "Upstream" ...
 $ chelae_length: num [1:180] 17 14.8 15.2 15.2 16 ...

```

```
summary(crayfish_data)
```

crayfish_id	river	position	chelae_length
Length:180	Length:180	Length:180	Min. : 5.985
Class :character	Class :character	Class :character	1st Qu.:10.208

```

Mode :character Mode :character Mode :character Median :13.877
Mean :13.727
3rd Qu.:17.247
Max. :22.336

```

```

# Convert position to factor with meaningful order
crayfish_data$position <- factor(crayfish_data$position,
                                levels = c("Upstream", "Downstream"))

head(crayfish_data, 10)

```

```

# A tibble: 10 × 4
  crayfish_id      river position chelae_length
  <chr>          <chr>   <fct>         <dbl>
1 River_A_Upstream_1 River_A Upstream      17.0
2 River_A_Upstream_2 River_A Upstream      14.8
3 River_A_Upstream_3 River_A Upstream      15.2
4 River_A_Upstream_4 River_A Upstream      15.2
5 River_A_Upstream_5 River_A Upstream      16.0
6 River_A_Upstream_6 River_A Upstream      14.6
7 River_A_Upstream_7 River_A Upstream      16.3
8 River_A_Upstream_8 River_A Upstream      14.5
9 River_A_Upstream_9 River_A Upstream      15.7
10 River_A_Upstream_10 River_A Upstream      16.4

```

Exploratory Data Analysis

```

# Basic summary statistics
summary_stats <- crayfish_data %>%
  group_by(river, position) %>%
  summarise(
    n = n(),
    mean_chelae = mean(chelae_length),
    sd_chelae = sd(chelae_length),
    se_chelae = sd_chelae / sqrt(n),
    .groups = "drop"
  )

print(summary_stats)

```

```

# A tibble: 12 × 6
  river position      n mean_chelae sd_chelae se_chelae
  <chr>   <fct>   <int>      <dbl>    <dbl>    <dbl>
1 River_A Upstream    15      15.5     1.03     0.266
2 River_A Downstream  15      18.3     1.75     0.452
3 River_B Upstream    15       8.04     1.10     0.285
4 River_B Downstream  15      10.6     0.916     0.236
5 River_C Upstream    15      18.0     1.20     0.310
6 River_C Downstream  15      20.5     1.24     0.319
7 River_D Upstream    15       8.16     1.37     0.354
8 River_D Downstream  15      10.7     1.38     0.357
9 River_E Upstream    15      14.0     1.15     0.296
10 River_E Downstream  15      17.2     1.43     0.370

```

11	River_F Upstream	15	10.5	1.25	0.323
12	River_F Downstream	15	13.2	1.15	0.296

```
# Overall means by position (ignoring river structure)
overall_means <- crayfish_data %>%
  group_by(position) %>%
  summarise(
    n = n(),
    mean_chelae = mean(chelae_length),
    sd_chelae = sd(chelae_length),
    .groups = "drop"
  )

print("Overall means by position (ignoring river structure):")
```

```
[1] "Overall means by position (ignoring river structure):"
```

```
print(overall_means)
```

```
# A tibble: 2 × 4
  position      n mean_chelae sd_chelae
  <fct>    <int>    <dbl>    <dbl>
1 Upstream    90     12.4     3.93
2 Downstream  90     15.1     4.06
```

```
# Means by river (ignoring position)
river_means <- crayfish_data %>%
  group_by(river) %>%
  summarise(
    n = n(),
    mean_chelae = mean(chelae_length),
    sd_chelae = sd(chelae_length),
    .groups = "drop"
  )

print("Means by river (showing extreme variation):")
```

```
[1] "Means by river (showing extreme variation):"
```

```
print(river_means)
```

```
# A tibble: 6 × 4
  river      n mean_chelae sd_chelae
  <chr>  <int>    <dbl>    <dbl>
1 River_A  30     16.9     2.02
2 River_B  30      9.33     1.65
3 River_C  30     19.3     1.76
4 River_D  30      9.41     1.86
5 River_E  30     15.6     2.06
6 River_F  30     11.9     1.79
```

```

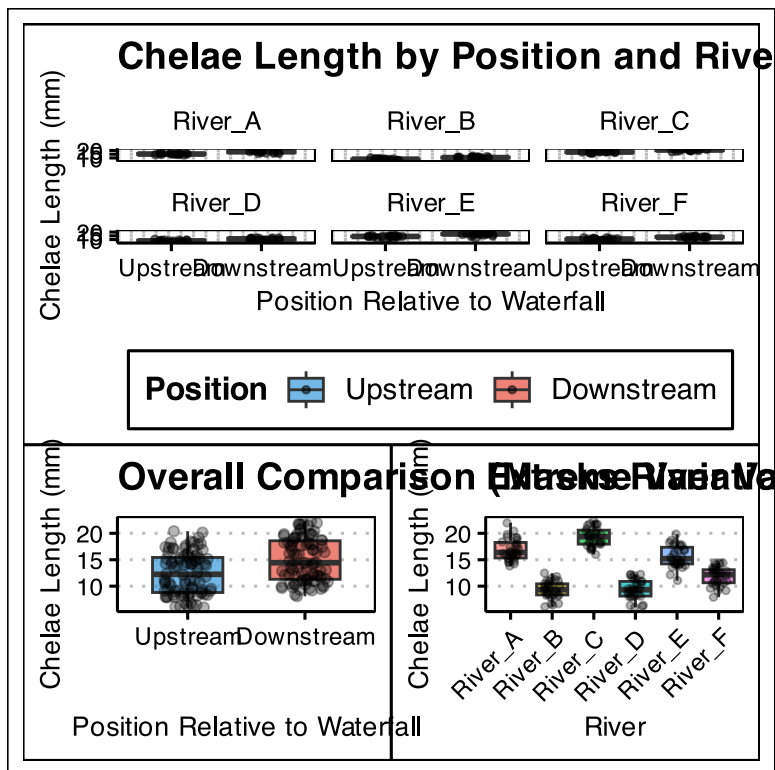
# Boxplot by position and river - shows extreme river variation
p1 <- ggplot(crayfish_data, aes(x = position, y = chelae_length, fill = position)) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(width = 0.2, alpha = 0.5, size = 1) +
  facet_wrap(~river, ncol = 3) +
  scale_fill_manual(values = c("Upstream" = "#3498db", "Downstream" = "#e74c3c")) +
  labs(title = "Chelae Length by Position and River (Note Extreme River Variation)",
       x = "Position Relative to Waterfall",
       y = "Chelae Length (mm)",
       fill = "Position") +
  theme(legend.position = "bottom")

# Overall comparison - this masks the river effects
p2 <- ggplot(crayfish_data, aes(x = position, y = chelae_length, fill = position)) +
  geom_boxplot(alpha = 0.7, width = 0.6) +
  geom_jitter(width = 0.2, alpha = 0.3, size = 1.5) +
  scale_fill_manual(values = c("Upstream" = "#3498db", "Downstream" = "#e74c3c")) +
  labs(title = "Overall Comparison (Masks River Variation)",
       x = "Position Relative to Waterfall",
       y = "Chelae Length (mm)",
       fill = "Position") +
  theme(legend.position = "none")

# Show river variation
p3 <- ggplot(crayfish_data, aes(x = river, y = chelae_length, fill = river)) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(width = 0.2, alpha = 0.3, size = 1) +
  labs(title = "Extreme Variation Between Rivers",
       x = "River",
       y = "Chelae Length (mm)") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")

# Combine plots
(p1) / (p2 + p3)

```



Create Averaged Data for Comparison

```
# Create averaged data (what many researchers incorrectly do)
averaged_data <- crayfish_data %>%
  group_by(river, position) %>%
  summarise(
    mean_chelae = mean(chelae_length),
    .groups = "drop"
  )

print("Averaged data (6 rivers x 2 positions = 12 data points):")
```

```
[1] "Averaged data (6 rivers x 2 positions = 12 data points):"
```

```
print(averaged_data)
```

```
# A tibble: 12 x 3
  river position mean_chelae
<chr> <fct>      <dbl>
1 River_A Upstream    15.5
2 River_A Downstream  18.3
3 River_B Upstream     8.04
4 River_B Downstream  10.6
5 River_C Upstream    18.0
6 River_C Downstream  20.5
7 River_D Upstream     8.16
8 River_D Downstream  10.7
9 River_E Upstream    14.0
10 River_E Downstream  17.2
```

```
11 River_F Upstream      10.5
12 River_F Downstream    13.2
```

```
# Show the dramatic loss of information
cat("Original data: ", nrow(crayfish_data), " observations\n")
```

```
Original data: 180 observations
```

```
cat("Averaged data: ", nrow(averaged_data), " observations\n")
```

```
Averaged data: 12 observations
```

```
cat("Information loss: ", round((1 - nrow(averaged_data)/nrow(crayfish_data)) * 100, 1), "%\n")
```

```
Information loss: 93.3 %
```

Method 1: INCORRECT - One-way ANOVA on Averaged Data

```
cat("=== INCORRECT METHOD: One-way ANOVA on averaged data ===\n\n")
```

```
=== INCORRECT METHOD: One-way ANOVA on averaged data ===
```

```
# Fit one-way ANOVA on averaged data (WRONG!)
anova_averaged <- aov(mean_chelae ~ position, data = averaged_data)
summary_averaged <- summary(anova_averaged)

print("One-way ANOVA on averaged data:")
```

```
[1] "One-way ANOVA on averaged data:"
```

```
print(summary_averaged)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
position	1	22.09	22.09	1.288	0.283
Residuals	10	171.49	17.15		

```
# Extract statistics
f_stat_avg <- summary_averaged[[1]]$`F value`[1]
p_value_avg <- summary_averaged[[1]]$`Pr(>F)`[1]

cat("Results from averaged data:\n")
```

```
Results from averaged data:
```



```
cat("F(1,10) =", round(f_stat_avg, 3), "\n")
```

F(1,10) = 1.288

```
cat("p-value =", round(p_value_avg, 3), "\n")
```

p-value = 0.283

```
cat("Conclusion:", ifelse(p_value_avg < 0.05, "Significant", "Not significant"), "\n\n")
```

Conclusion: Not significant

```
# Problems with this approach  
cat("PROBLEMS WITH THIS APPROACH:\n")
```

PROBLEMS WITH THIS APPROACH:

```
cat("1. Treats rivers as replicates (pseudoreplication)\n")
```

1. Treats rivers as replicates (pseudoreplication)

```
cat("2. Ignores within-river correlation\n")
```

2. Ignores within-river correlation

```
cat("3. Loses individual-level information\n")
```

3. Loses individual-level information

```
cat("4. Inflated Type I error rate\n")
```

4. Inflated Type I error rate

```
cat("5. Wrong degrees of freedom\n\n")
```

5. Wrong degrees of freedom

Method 2: CORRECT - Mixed-Effects Model

```
cat("=== CORRECT METHOD: Mixed-effects model ===\n\n")
```

```
=== CORRECT METHOD: Mixed-effects model ===
```

```
# Fit the mixed-effects model
model <- lmer(chelae_length ~ position + (1|river), data = crayfish_data)

# Model summary
summary(model)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: chelae_length ~ position + (1 | river)
Data: crayfish_data
```

```
REML criterion at convergence: 623.3
```

```
Scaled residuals:
```

Min	1Q	Median	3Q	Max
-2.54215	-0.64713	0.09117	0.69890	2.97664

```
Random effects:
```

Groups	Name	Variance	Std.Dev.
river	(Intercept)	17.080	4.133
Residual		1.569	1.253

```
Number of obs: 180, groups: river, 6
```

```
Fixed effects:
```

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	12.3700	1.6923	5.0306	7.309	0.000731 ***
positionDownstream	2.7137	0.1867	173.0000	14.531	< 2e-16 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Correlation of Fixed Effects:
```

```
(Intr)
pstinDwnstrm -0.055
```

```
# ANOVA with proper F-test
anova_mixed <- anova(model)
print("Mixed-effects ANOVA:")
```

```
[1] "Mixed-effects ANOVA:"
```

```
print(anova_mixed)
```

```
Type III Analysis of Variance Table with Satterthwaite's method
```

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
position	331.38	331.38	1	173	211.16	< 2.2e-16 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Extract statistics
f_stat_mixed <- anova_mixed$`F value`[1]
p_value_mixed <- anova_mixed$`Pr(>F)`[1]
df_num <- anova_mixed$NumDF[1]
df_den <- round(anova_mixed$DenDF[1], 1)

cat("Results from mixed-effects model:\n")
```

Results from mixed-effects model:

```
cat("F(", df_num, ",", df_den, ") =", round(f_stat_mixed, 3), "\n")
```

F(1 , 173) = 211.161

```
cat("p-value =", round(p_value_mixed, 3), "\n")
```

p-value = 0

```
cat("Conclusion:", ifelse(p_value_mixed < 0.05, "Significant", "Not significant"), "\n\n")
```

Conclusion: Significant

```
# Variance components
var_components <- as.data.frame(VarCorr(model))
river_var <- var_components$vcov[1]
residual_var <- var_components$vcov[2]
icc <- river_var / (river_var + residual_var)

cat("Variance components:\n")
```

Variance components:

```
cat("River variance:", round(river_var, 3), "\n")
```

River variance: 17.08

```
cat("Residual variance:", round(residual_var, 3), "\n")
```

Residual variance: 1.569

```
cat("ICC (river effect):", round(icc, 3), "\n")
```

ICC (river effect): 0.916

```
cat("This means", round(icc*100, 1), "% of variance is between rivers\n\n")
```

This means 91.6 % of variance is between rivers

Direct Comparison of Methods

```
cat("=== DIRECT COMPARISON OF METHODS ===\n\n")
```

```
=== DIRECT COMPARISON OF METHODS ===
```

```
# Create comparison table
comparison <- data.frame(
  Method = c("One-way ANOVA (averaged)", "Mixed-effects model"),
  F_statistic = c(round(f_stat_avg, 3), round(f_stat_mixed, 3)),
  DF = c("1, 10", paste0(df_num, ", ", df_den)),
  P_value = c(round(p_value_avg, 3), round(p_value_mixed, 3)),
  Significant = c(ifelse(p_value_avg < 0.05, "Yes", "No"),
                 ifelse(p_value_mixed < 0.05, "Yes", "No")),
  N_observations = c(nrow(averaged_data), nrow(crayfish_data))
)

print(comparison)
```

	Method	F_statistic	DF	P_value	Significant
1	One-way ANOVA (averaged)	1.288	1, 10	0.283	No
2	Mixed-effects model	211.161	1, 173	0.000	Yes
N_observations					
1		12			
2		180			

```
cat("\nKEY DIFFERENCES:\n")
```

KEY DIFFERENCES:

```
cat("1. Degrees of freedom: ANOVA uses", 10, "error df, mixed model uses", round(df_den, 1), "\n")
```

1. Degrees of freedom: ANOVA uses 10 error df, mixed model uses 173

```
cat("2. Sample size: ANOVA uses", nrow(averaged_data), "points, mixed model uses",
nrow(crayfish_data), "\n")
```

2. Sample size: ANOVA uses 12 points, mixed model uses 180

```
cat("3. Power: Mixed model has", ifelse(p_value_mixed < p_value_avg, "higher", "lower"), "power\n")
```

3. Power: Mixed model has higher power

```
cat("4. Validity: Only mixed model accounts for river clustering\n\n")
```

4. Validity: Only mixed model accounts for river clustering

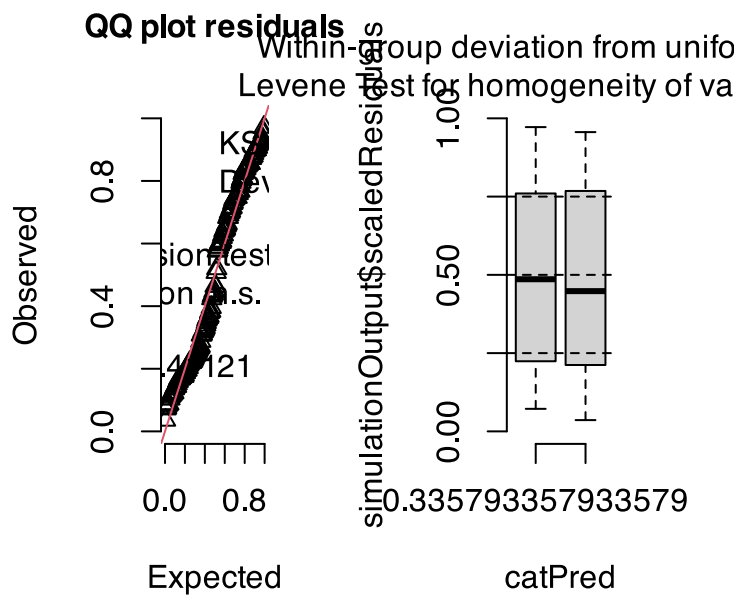
Model Assumptions Testing

```
# DHARMA residual diagnostics for mixed model
simulationOutput <- simulateResiduals(fittedModel = model, plot = FALSE)

# Create assumption plots
par(mfrow = c(2, 2))

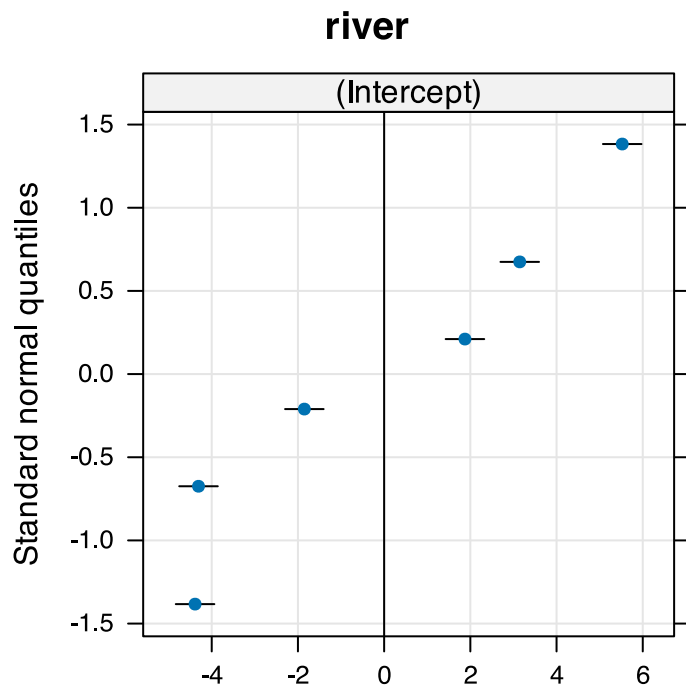
# 1. DHARMA residual plot
plot(simulationOutput, main = "DHARMA Residual Diagnostics")
```

DHARMA residual



```
# 2. QQ plot for random effects
print(lattice::qqmath(ranef(model)))
```

```
$river
```



```
par(mfrow = c(1, 1))  
  
# Formal tests  
print("Shapiro-Wilk test for residuals:")
```

```
[1] "Shapiro-Wilk test for residuals:"
```

```
shapiro_test <- shapiro.test(residuals(model))  
print(shapiro_test)
```

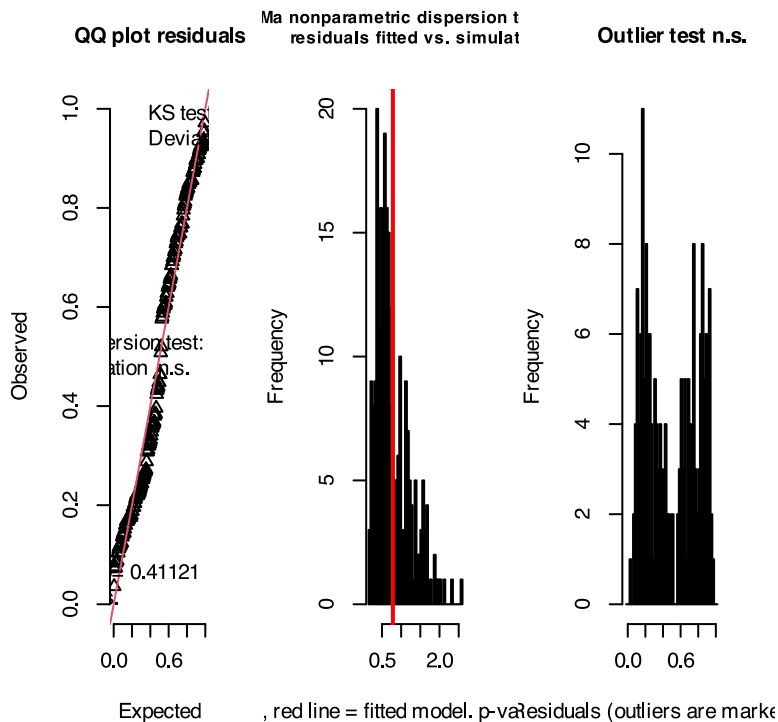
Shapiro-Wilk normality test

```
data: residuals(model)  
W = 0.99219, p-value = 0.4436
```

```
print("DHARMA tests:")
```

```
[1] "DHARMA tests:"
```

```
testResiduals(simulationOutput)
```



```
$uniformity
```

Asymptotic one-sample Kolmogorov-Smirnov test

```
data: simulationOutput$scaledResiduals
```

```
D = 0.083556, p-value = 0.1619
```

```
alternative hypothesis: two-sided
```

```
$dispersion
```

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

```
data: simulationOutput
```

```
dispersion = 1.0001, p-value = 0.744
```

```
alternative hypothesis: two.sided
```

```
$outliers
```

DHARMA outlier test based on exact binomial test with approximate expectations

```
data: simulationOutput
```

```
outliers at both margin(s) = 0, observations = 180, p-value = 0.4112
```

```
alternative hypothesis: true probability of success is not equal to 0.007968127
```

```
95 percent confidence interval:
```

```
0.0000000 0.0202852
```

```
sample estimates:
```

```
frequency of outliers (expected: 0.00796812749003984 )
```

```
0
```

```

# Create residual plots using ggplot
residual_data <- data.frame(
  fitted = fitted(model),
  residuals = residuals(model),
  river = crayfish_data$river,
  position = crayfish_data$position
)

# QQ plot for random effects using ggplot
ranef_data <- ranef(model)$river
ranef_df <- data.frame(
  river = rownames(ranef_data),
  intercept = ranef_data[,1]
)

p_qq_ranef <- ggplot(ranef_df, aes(sample = intercept)) +
  stat_qq() +
  stat_qq_line() +
  labs(title = "QQ Plot: Random Effects (River Intercepts)",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  theme_minimal()

# Residuals vs Fitted
p_resid1 <- ggplot(residual_data, aes(x = fitted, y = residuals)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "loess", se = FALSE, color = "red") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(title = "Residuals vs Fitted Values",
       x = "Fitted Values", y = "Residuals")

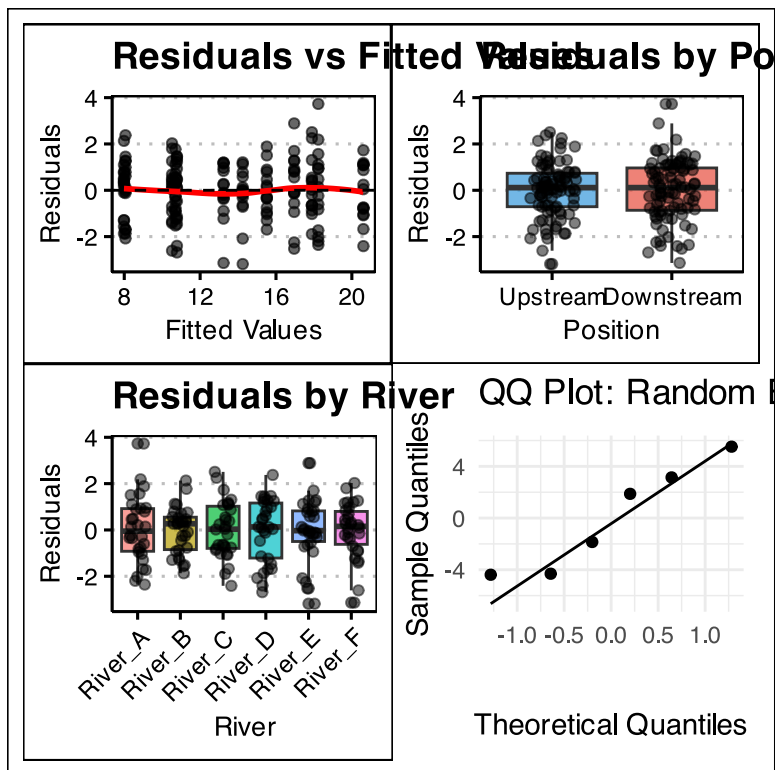
# Residuals by group
p_resid2 <- ggplot(residual_data, aes(x = position, y = residuals, fill = position)) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(width = 0.2, alpha = 0.5) +
  scale_fill_manual(values = c("Upstream" = "#3498db", "Downstream" = "#e74c3c")) +
  labs(title = "Residuals by Position",
       x = "Position", y = "Residuals") +
  theme(legend.position = "none")

# Residuals by river
p_resid3 <- ggplot(residual_data, aes(x = river, y = residuals, fill = river)) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(width = 0.2, alpha = 0.5) +
  labs(title = "Residuals by River",
       x = "River", y = "Residuals") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
       legend.position = "none")

# Combine assumption plots
(p_resid1 + p_resid2) / (p_resid3 + p_qq_ranef)

```

```
`geom_smooth()` using formula = 'y ~ x'
```

Post-hoc Analysis and Effect Sizes

```
# Estimated marginal means
emm <- emmeans(model, ~ position)
print("Estimated marginal means:")
```

```
[1] "Estimated marginal means:"
```

```
print(emm)
```

```
position  emmean   SE   df lower.CL upper.CL
Upstream    12.4 1.69 5.03     8.03    16.7
Downstream   15.1 1.69 5.03    10.74    19.4
```

```
Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95
```

```
# Pairwise contrasts
contrasts <- pairs(emm)
print("Pairwise contrasts:")
```

```
[1] "Pairwise contrasts:"
```

```
print(contrasts)
```

```
contrast          estimate    SE   df t.ratio p.value
Upstream - Downstream   -2.71 0.187 173  -14.531  <.0001
```

Degrees-of-freedom method: kenward-roger

```
# Effect size calculation (Cohen's d for mixed models)
means_by_pos <- crayfish_data %>%
  group_by(position) %>%
  summarise(mean_chelae = mean(chelae_length), .groups = "drop")

# Calculate Cohen's d using the residual SD from the model
residual_sd <- sigma(model)
cohens_d <- diff(means_by_pos$mean_chelae) / residual_sd

cat("Cohen's d effect size:", round(cohens_d, 3), "\n")
```

Cohen's d effect size: 2.166

```
cat("Interpretation:",
    ifelse(abs(cohens_d) < 0.2, "negligible",
           ifelse(abs(cohens_d) < 0.5, "small",
                  ifelse(abs(cohens_d) < 0.8, "medium", "large"))), "effect\n")
```

Interpretation: large effect

```
# Confidence intervals for fixed effects
confint_results <- confint(model, method = "Wald")
print("Confidence intervals:")
```

[1] "Confidence intervals:"

```
print(confint_results)
```

	2.5 %	97.5 %
.sig01	NA	NA
.sigma	NA	NA
(Intercept)	9.053097	15.686955
positionDownstream	2.347655	3.079683

Visualization Comparing Both Approaches

```
# Create plots showing why mixed model is better

# Plot 1: Raw data with river structure
p_raw <- ggplot(crayfish_data, aes(x = position, y = chelae_length)) +
  geom_point(aes(color = river), alpha = 0.7, size = 2,
             position = position_jitter(width = 0.2)) +
  geom_line(aes(group = river, color = river),
            stat = "summary", fun = "mean", size = 1.2, alpha = 0.8) +
  stat_summary(fun = "mean", geom = "point", size = 4, shape = 21,
              fill = "white", stroke = 2) +
```

```

stat_summary(fun.data = "mean_se", geom = "errorbar",
             width = 0.1, size = 1.2) +
labs(title = "Mixed Model Approach: Accounts for River Structure",
     subtitle = "Lines connect river means, black points show overall means  $\pm$  SE",
     x = "Position", y = "Chelae Length (mm)") +
theme(legend.position = "right")

```

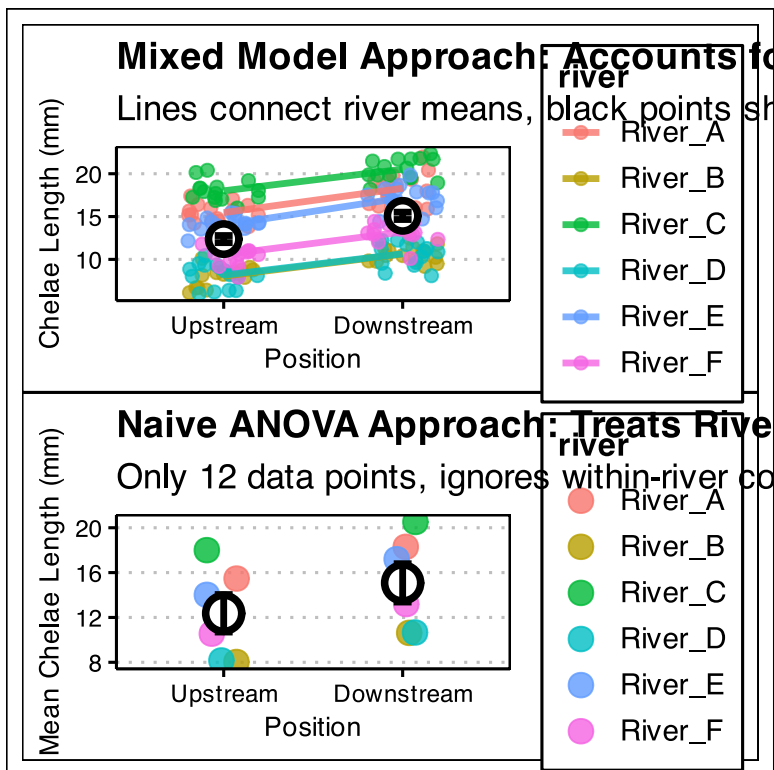
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 i Please use `linewidth` instead.

```

# Plot 2: Averaged data approach
p_averaged <- ggplot(averaged_data, aes(x = position, y = mean_chelae)) +
  geom_point(aes(color = river), size = 4, alpha = 0.8,
            position = position_jitter(width = 0.1)) +
  stat_summary(fun = "mean", geom = "point", size = 5, shape = 21,
              fill = "white", stroke = 2) +
  stat_summary(fun.data = "mean_se", geom = "errorbar",
              width = 0.1, size = 1.2) +
  labs(title = "Naive ANOVA Approach: Treats Rivers as Replicates",
       subtitle = "Only 12 data points, ignores within-river correlation",
       x = "Position", y = "Mean Chelae Length (mm)") +
  theme(legend.position = "right")

# Combine plots
p_raw / p_averaged

```



Publication-Quality Figure

```

# Create summary data for error bars using mixed model predictions
plot_data <- crayfish_data %>%

```

```

group_by(river, position) %>%
  summarise(
    mean_chelae = mean(chelae_length),
    se_chelae = sd(chelae_length) / sqrt(n()),
    .groups = "drop"
  )

# Create the main figure
final_plot <- ggplot(plot_data, aes(x = position, y = mean_chelae, fill = position)) +
  # Add individual points (raw data) with transparency
  geom_jitter(data = crayfish_data,
    aes(x = position, y = chelae_length),
    width = 0.25, alpha = 0.3, size = 1.2,
    inherit.aes = FALSE) +

  # Add means with error bars
  geom_col(alpha = 0.7, width = 0.6, color = "black", size = 0.5) +
  geom_errorbar(aes(ymin = mean_chelae - se_chelae,
    ymax = mean_chelae + se_chelae),
    width = 0.2, size = 0.8) +

  # Facet by river
  facet_wrap(~river, ncol = 3,
    labeller = labeller(river = function(x) gsub("_", " ", x))) +

  # Customize colors and theme
  scale_fill_manual(values = c("Upstream" = "#2c3e50", "Downstream" = "#c0392b"),
    name = "Position") +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1))) +

  # Labels and theme
  labs(
    title = "Crayfish Chelae Length: Extreme River Variation Requires Mixed Models",
    subtitle = "Mean ± SE shown with individual observations; note massive between-river differences",
    x = "Position Relative to Waterfall",
    y = "Chelae Length (mm)",
    caption = paste0("Mixed-effects model: F(", df_num, ", ", round(df_den, 1), ") = ",
      round(f_stat_mixed, 2), ", p = ",
      ifelse(p_value_mixed < 0.001, "< 0.001", round(p_value_mixed, 3)))
  ) +

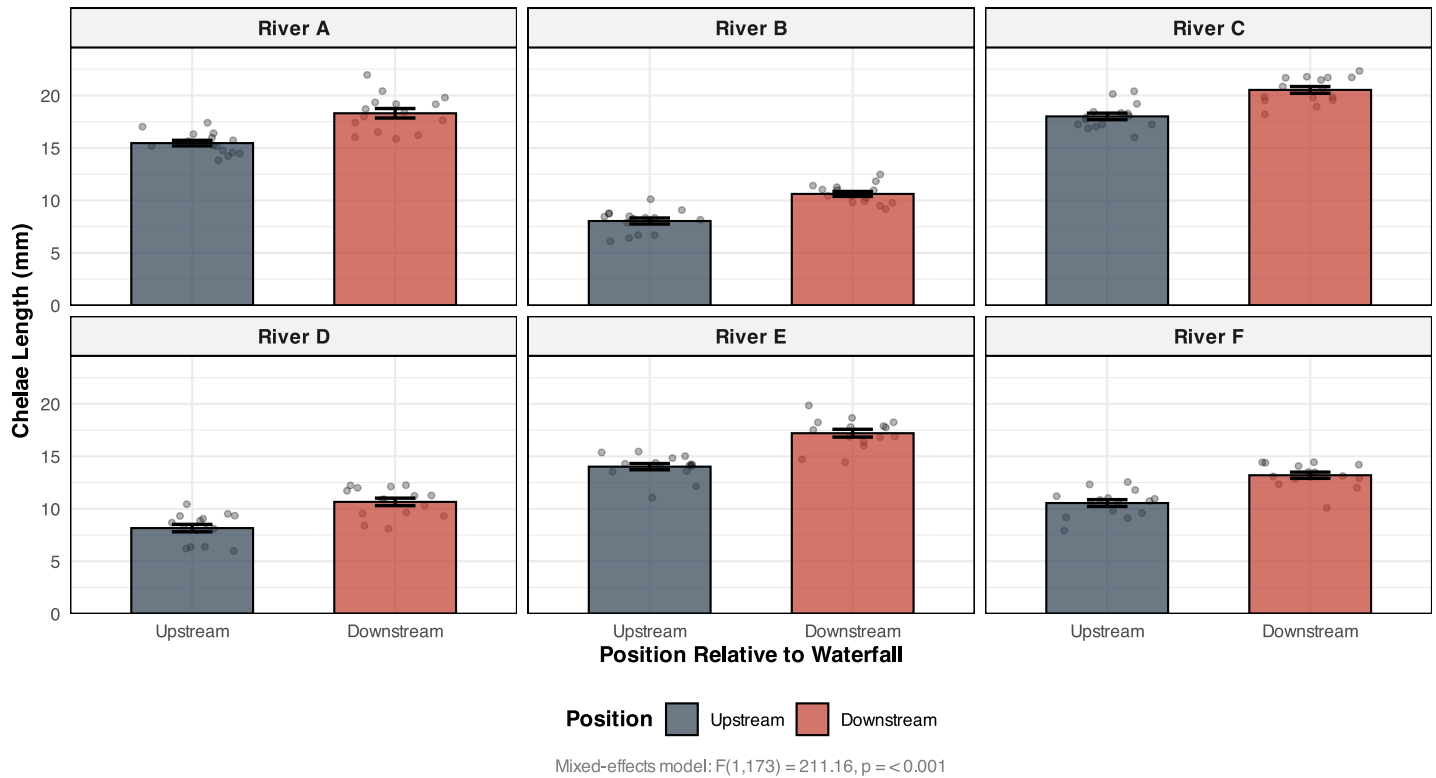
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    plot.subtitle = element_text(size = 11, hjust = 0.5, color = "gray40"),
    strip.background = element_rect(fill = "gray95", color = "black"),
    strip.text = element_text(face = "bold", size = 10),
    panel.border = element_rect(color = "black", fill = NA),
    legend.position = "bottom",
    legend.title = element_text(face = "bold"),
    axis.title = element_text(face = "bold"),
    plot.caption = element_text(hjust = 0.5, color = "gray50")
  )

print(final_plot)

```

Crayfish Chelae Length: Extreme River Variation Requires Mixed Models

Mean $\pm$ SE shown with individual observations; note massive between-river differences



```
# Save the plot
ggsave("crayfish_chelae_mixed_model.png", final_plot,
       width = 10, height = 6, dpi = 300, bg = "white")
```

Summary of Why Mixed Models Are Superior

```
cat("=== WHY MIXED-EFFECTS MODELS ARE SUPERIOR ===\n\n")
```

```
=== WHY MIXED-EFFECTS MODELS ARE SUPERIOR ===
```

```
cat("1. STATISTICAL VALIDITY:\n")
```

```
1. STATISTICAL VALIDITY:
```

```
cat("    • Accounts for non-independence within rivers\n")
```

```
    • Accounts for non-independence within rivers
```

```
cat("    • Correct degrees of freedom (", round(df_den,1), " vs 10)\n")
```

```
    • Correct degrees of freedom ( 173 vs 10)
```

```
cat("    • Proper error structure\n\n")
```

- Proper error structure

```
cat("2. POWER AND PRECISION:\n")
```

2. POWER AND PRECISION:

```
cat("    • Uses all", nrow(crayfish_data), "observations vs", nrow(averaged_data), "\n")
```

- Uses all 180 observations vs 12

```
cat("    • Better estimates of treatment effects\n")
```

- Better estimates of treatment effects

```
cat("    • More appropriate confidence intervals\n\n")
```

- More appropriate confidence intervals

```
cat("3. BIOLOGICAL REALISM:\n")
```

3. BIOLOGICAL REALISM:

```
cat("    • Separates treatment effects from river effects\n")
```

- Separates treatment effects from river effects

```
cat("    • ICC =", round(icc, 3), "shows", round(icc*100,1), "% of variation is between rivers\n")
```

- ICC = 0.916 shows 91.6 % of variation is between rivers

```
cat("    • Accounts for natural hierarchical structure\n\n")
```

- Accounts for natural hierarchical structure

```
cat("4. FLEXIBILITY:\n")
```

4. FLEXIBILITY:

```
cat("    • Can handle unbalanced designs\n")
```

- Can handle unbalanced designs

```
cat("    • Easily extended to include covariates\n")
```

- Easily extended to include covariates

```
cat("    • Can model complex random effect structures\n\n")
```

- Can model complex random effect structures

```
river_range <- max(river_means$mean_chelae) - min(river_means$mean_chelae)
treatment_effect <- diff(means_by_pos$mean_chelae)
```

```
cat("5. IN THIS DATASET:\n")
```

```
5. IN THIS DATASET:
```

```
cat("    • River variation range:", round(river_range, 2), "mm\n")
```

- River variation range: 9.94 mm

```
cat("    • Treatment effect:", round(treatment_effect, 2), "mm\n")
```

- Treatment effect: 2.71 mm

```
cat("    • River variation is", round(river_range/treatment_effect, 1), "× larger than treatment effect!\n")
```

- River variation is 3.7 × larger than treatment effect!

```
cat("    • This makes accounting for river effects crucial\n\n")
```

- This makes accounting for river effects crucial

```
cat("CONCLUSION: Mixed-effects models are essential when data have hierarchical structure!\n")
```

```
CONCLUSION: Mixed-effects models are essential when data have hierarchical structure!
```

Session Information

```
sessionInfo()
```

```
R version 4.5.1 (2025-06-13)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.6

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK
version 3.12.1

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Chicago
tzcode source: internal

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods   base

other attached packages:
 [1] lmerTest_3.1-3      patchwork_1.3.1      ggthemes_5.1.0
 [4] broom.mixed_0.2.9.6 emmeans_1.11.2       DHARMa_0.4.7
 [7] car_3.1-3           carData_3.0-5        lme4_1.1-37
[10] Matrix_1.7-3        lubridate_1.9.4      forcats_1.0.0
[13] stringr_1.5.1       dplyr_1.1.4          purrr_1.1.0
[16] readr_2.1.5         tidyr_1.3.1          tibble_3.3.0
[19] ggplot2_3.5.2       tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] tidyselect_1.2.1    farver_2.1.2         fastmap_1.2.0
 [4] TH.data_1.1-3       digest_0.6.37        estimability_1.5.1
 [7] timechange_0.3.0    lifecycle_1.0.4      survival_3.8-3
[10] magrittr_2.0.3      compiler_4.5.1       rlang_1.1.6
[13] tools_4.5.1         utf8_1.2.6           yaml_2.3.10
[16] knitr_1.50          labeling_0.4.3       RColorBrewer_1.1-3
[19] gap.datasets_0.0.6  multcomp_1.4-28      abind_1.4-8
[22] withr_3.0.2         numDeriv_2016.8-1.1 grid_4.5.1
[25] xtable_1.8-4        future_1.67.0        globals_0.18.0
[28] scales_1.4.0        MASS_7.3-65          cli_3.6.5
[31] mvtnorm_1.3-3       rmarkdown_2.29       ragg_1.4.0
[34] reformulas_0.4.1    generics_0.1.4       rstudioapi_0.17.1
[37] tzdb_0.5.0          minqa_1.2.8          splines_4.5.1
[40] parallel_4.5.1      vctrs_0.6.5          boot_1.3-31
[43] sandwich_3.1-1     jsonlite_2.0.0       hms_1.1.3
[46] pbkrtest_0.5.5      Formula_1.2-5        listenv_0.9.1
[49] systemfonts_1.2.3   gap_1.6              glue_1.8.0
[52] parallelly_1.45.1  nloptr_2.2.1         codetools_0.2-20
[55] stringi_1.8.7       gtable_0.3.6         pillar_1.11.0
[58] furrr_0.3.1         htmltools_0.5.8.1    R6_2.6.1
[61] textshaping_1.0.1   Rdpack_2.6.4         evaluate_1.0.4
[64] lattice_0.22-7      rbibutils_2.3         backports_1.5.0
[67] broom_1.0.9         Rcpp_1.1.0           coda_0.19-4.1
[70] nlme_3.1-168        mgcv_1.9-3           xfun_0.52
[73] zoo_1.8-14          pkgconfig_2.0.3
```